

GASTONIA, NC, USA

WMO: 723147

Lat: 35.197N

Lon: 81.156W

Elev: 797

StdP: 14.28

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 53870

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.6	24.8	1.3	6.0	29.4	6.0	7.7	33.1	16.8	47.9	15.1	44.8	2.6	0	0.362

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	94.6	74.1	92.4	74.1	90.3	73.7	77.4	88.7	76.6	87.4	75.6	85.8	4.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.3	131.8	81.7	73.3	127.4	80.8	72.6	124.3	80.1	41.2	88.9	40.4	87.7	39.5	86.4	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.8	12.4	10.9	13.5	97.4	4.4	4.2	10.4	100.4	7.8	102.8	5.4	105.2	2.2	108.2		
			11.6	79.5	4.0	1.1	8.8	80.3	6.5	80.9	4.3	81.5	1.4	82.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	61.7	42.1	45.6	53.0	61.6	69.2	76.7	79.4	78.6	73.2	62.2	51.9	46.0
	DBStd	15.04	10.04	9.16	9.17	7.40	6.19	4.23	3.50	4.12	5.63	7.76	8.03	8.81
	HDD50	816	287	181	76	7	0	0	0	0	0	6	71	188
	HDD65	2965	710	547	383	148	35	0	0	0	9	146	396	591
	CDD50	5087	43	56	169	355	594	801	913	886	695	383	129	63
	CDD65	1763	0	2	12	46	165	351	448	421	254	58	5	1
	CDH74	15555	2	12	122	500	1384	3193	4189	3743	1933	439	32	6
	CDH80	6327	0	0	16	100	420	1366	1930	1664	731	99	1	0
Wind	WSAvg	4.1	4.5	4.9	5.2	4.8	3.9	3.3	3.1	3.1	4.1	3.8	4.1	4.0
Precipitation	PrecAvg	42.2	3.0	2.9	4.0	3.3	3.7	4.1	3.8	3.9	3.8	2.6	3.1	3.8
	PrecMax	57.6	5.1	4.3	7.4	8.5	9.5	8.3	7.6	9.4	7.6	5.4	9.3	8.5
	PrecMin	28.1	1.0	1.2	1.2	0.9	0.5	1.2	1.5	0.7	0.3	0.0	0.5	1.0
	PrecStd	7.9	1.3	1.0	1.7	2.1	2.4	2.1	1.7	2.2	1.7	1.6	2.3	2.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.0	75.9	82.2	86.3	91.7	96.9	97.4	98.1	94.0	87.7	78.7	73.5
		MCWB	60.5	62.9	61.9	65.6	71.0	73.6	75.1	75.1	71.3	72.2	63.9	62.2
	2%	DB	66.8	70.2	77.3	82.8	87.6	93.0	94.2	93.8	90.9	82.2	73.2	68.6
		MCWB	58.2	57.5	60.4	64.5	69.9	73.7	75.0	74.5	71.5	68.2	61.1	60.6
	5%	DB	63.1	66.0	73.2	80.1	84.4	90.3	92.0	91.2	87.8	79.1	70.0	64.9
		MCWB	55.9	55.7	58.5	63.2	69.3	73.5	75.0	74.6	71.4	65.9	60.0	58.5
	10%	DB	58.5	61.8	69.0	76.4	81.7	87.7	89.8	88.6	84.2	75.1	66.3	61.1
		MCWB	51.4	51.9	56.1	61.7	68.0	72.3	74.5	74.1	70.8	65.5	58.0	54.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.9	65.0	65.3	70.6	74.7	77.4	78.7	78.5	76.8	75.2	68.6	66.4
		MCDB	68.1	72.2	76.2	81.3	84.1	88.6	89.6	89.7	86.4	81.3	71.2	69.7
	2%	WB	60.4	61.8	63.1	68.2	72.7	76.1	77.5	77.4	75.1	72.3	65.3	63.3
		MCDB	65.2	66.8	72.4	77.1	82.3	87.6	89.2	88.5	84.0	78.2	69.9	67.1
	5%	WB	57.0	57.6	61.2	66.0	71.2	75.1	76.7	76.5	73.7	69.6	62.9	59.8
		MCDB	61.9	63.8	70.0	74.6	80.8	86.4	87.9	86.9	81.4	75.6	67.5	63.9
	10%	WB	52.5	53.2	59.0	64.0	69.9	74.1	75.7	75.4	72.4	67.0	59.5	55.6
		MCDB	57.3	59.3	66.3	72.6	78.6	84.6	86.2	85.0	79.9	73.3	65.2	59.7
Mean Daily Temperature Range	5% DB	MDBR	19.8	20.4	21.7	22.6	20.9	20.0	19.3	18.8	18.9	21.0	22.0	19.3
		MCDBR	23.7	24.8	26.1	26.6	24.1	23.8	22.6	22.6	23.6	23.9	25.4	21.7
		MCWBR	16.1	15.0	13.2	11.8	10.2	8.4	7.6	7.5	8.3	11.4	14.5	14.3
	5% WB	MCDBR	19.7	20.7	21.8	21.5	20.3	20.4	19.6	18.9	18.3	18.5	18.0	18.0
		MCWBR	15.5	15.5	13.0	10.9	9.3	8.3	7.6	7.5	7.8	10.2	13.3	14.2
Clear-Sky Solar Irradiance	taub	0.313	0.321	0.360	0.400	0.455	0.482	0.530	0.514	0.425	0.381	0.332	0.314	
	taud	2.498	2.469	2.396	2.277	2.161	2.125	2.015	2.045	2.279	2.381	2.481	2.527	
	Ebn at Noon	283	293	289	281	266	258	244	246	264	267	273	276	
	Edh at Noon	26	30	36	42	48	49	55	52	40	33	27	24	
All-Sky Solar Radiation	RadAvg	838	1044	1381	1723	1912	2029	1897	1741	1480	1220	925	720	
	RadStd	49	119	99	134	156	146	127	83	132	149	77	68	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+1.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (62 neighbors)	+0.94	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+292	+194

Nomenclature: See separate page